

CBSE Practice Worksheet

Class 6 | Science | Difficulty: Hard | Set E

Instructions: Answer all questions. This worksheet contains 25 questions including multiple choice, fill in the blanks, and short answer questions.

Section A: Multiple Choice Questions (1 mark each)

1. Which principle explains the working of a hydraulic lift?
(A) Archimedes
(B) Pascal
(C) Bernoulli
(D) Newton
2. Which enzyme is responsible for DNA replication?
(A) DNA ligase
(B) DNA polymerase
(C) Helicase
(D) Primase
3. What is the wavelength of de Broglie wave for an electron?
(A) $\lambda = h/mv$
(B) $\lambda = mv/h$
(C) $\lambda = hv/m$
(D) $\lambda = m/hv$
4. What is the magnetic quantum number for 3d orbital?
(A) -2 to +2
(B) -3 to +3
(C) -1 to +1
(D) 0 only
5. What is the coordination number in FCC structure?
(A) 6
(B) 8
(C) 12
(D) 14
6. What is the hybridization of carbon in methane?
(A) sp
(B) sp²
(C) sp³
(D) sp³d
7. What is the wavelength of Lyman series limit?
(A) 91.2 nm
(B) 121.6 nm
(C) 656.3 nm
(D) 364.6 nm
8. Which reaction is endothermic?
(A) Combustion
(B) Neutralization
(C) Photosynthesis

(D) Respiration

9. Which organelle is involved in programmed cell death?

- (A) Lysosome
- (B) Mitochondria
- (C) Peroxisome
- (D) Ribosome

10. Which type of mutation involves change in single nucleotide?

- (A) Deletion
- (B) Insertion
- (C) Point mutation
- (D) Frameshift

Section B: Fill in the Blanks (1 mark each)

- 11. The IUPAC name of $\text{CH}_3\text{CH}_2\text{CHO}$ is ____.
- 12. In $\text{S}_\text{N}2$ reactions, the rate depends on both ____ and nucleophile.
- 13. The band gap in semiconductors is typically ____ eV.
- 14. The process of splitting water during photosynthesis is called ____.
- 15. For a reversible process, entropy change of the universe is ____.
- 16. The magnetic field inside a solenoid is given by $B = \mu \blacksquare$ ____.
- 17. The Krebs cycle occurs in the ____ of mitochondria.
- 18. The uncertainty principle was given by ____.

Section C: Short Answer Questions (2 marks each)

- 19. Describe the mechanism of nerve impulse transmission.
- 20. Explain Le Chatelier's principle with examples.
- 21. Explain hybridization with examples of sp , sp^2 , and sp^3 .
- 22. Describe the light-dependent and light-independent reactions of photosynthesis.
- 23. Describe electromagnetic induction and derive Faraday's law.
- 24. Explain the mechanism of DNA replication.
- 25. Explain the structure and working of a nuclear reactor.

--- End of Worksheet ---

CBSE Practice Hub | General Practice Questions